

Experiment No: 8

MINI PROJECT 1

Date

Aim:

To import, clean, transform, inspect, analyze, and visualize the Tourism Dataset using Python, Pandas, NumPy, and Matplotlib, and to derive meaningful insights from visitor, rating, and revenue data.

Procedure:

1. Import the required Python libraries such as **Pandas, NumPy, and Matplotlib**.
2. Import the Tourism Dataset using `pd.read_csv()` and store it in a DataFrame.
3. Verify the successful import by checking the dataset shape and displaying the first few records.
4. Check the dataset for **missing/null values** using `isnull().sum()`.
5. Check for **duplicate records** using `duplicated().sum()`.
6. Remove duplicate records using `drop_duplicates()`.
7. Check the **data types** of all columns using `dtypes`.
8. Perform data transformation by creating a new **Revenue_per_Visitor** column using Revenue divided by Visitors.
9. Inspect the structure and information of the dataset using `info()`.
10. Perform **statistical analysis** using `describe()` to obtain count, mean, standard deviation, minimum, maximum, and percentile values.
11. Filter the dataset based on visitor count to identify tourism records with more than **500,000 visitors**.
12. Perform **correlation analysis** among Visitors, Rating, Revenue, and Revenue_per_Visitor.
13. Create a **bar chart** to visualize total visitors by tourism category.
14. Create a **histogram** to visualize the distribution of visitors.
15. Create a **scatter plot** to analyze the relationship between visitors and revenue.
16. Create **box, bar, and line plots** to visualize revenue distribution, average revenue by category, and revenue across tourism records.
17. Export the cleaned and transformed dataset into a new CSV file using `to_csv()`

Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
df = pd.read_csv(r"C:\Users\SHREE HARINI\Downloads\archive (1)\tourism_dataset.csv")
```

```
print(df.shape)
```

```
(5989, 7)
```

```
print(df.head())
```

	Location	Country	Category	Visitors	Rating	Revenue \
0	kuBZRkVsAR	India	Nature	948853	1.32	84388.38
1	aHKUXhjzTo	USA	Historical	813627	2.01	802625.60
2	dIrdYtJFTA	Brazil	Nature	508673	1.42	338777.11
3	DxmlzdGkHK	Brazil	Historical	623329	1.09	295183.60
4	WJCCQlepnz	France	Cultural	124867	1.43	547893.24

```
Accommodation_Available
```

0	Yes
1	No
2	Yes
3	Yes
4	No

```
df["Revenue_per_Visitor"] = df["Revenue"] / df["Visitors"]
```

```
print(df[["Visitors", "Revenue", "Revenue_per_Visitor"]].head())
```

	Visitors	Revenue	Revenue_per_Visitor
0	948853	84388.38	0.088937
1	813627	802625.60	0.986479
2	508673	338777.11	0.666002
3	623329	295183.60	0.473560
4	124867	547893.24	4.387815

```
print(df.head())
```

	Location	Country	Category	Visitors	Rating	Revenue \
0	kuBZRkVsAR	India	Nature	948853	1.32	84388.38
1	aHKUXhjzTo	USA	Historical	813627	2.01	802625.60
2	dIrdYtJFTA	Brazil	Nature	508673	1.42	338777.11
3	DxmlzdGkHK	Brazil	Historical	623329	1.09	295183.60
4	WJCCQlepnz	France	Cultural	124867	1.43	547893.24

```
Accommodation_Available Revenue_per_Visitor
```

0	Yes	0.088937
1	No	0.986479
2	Yes	0.666002
3	Yes	0.473560
4	No	4.387815

```
print(df.info())
```

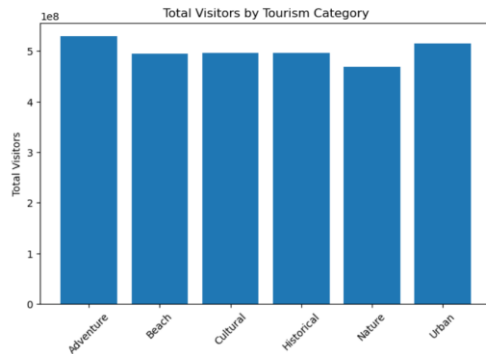
```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 5989 entries, 0 to 5988
```

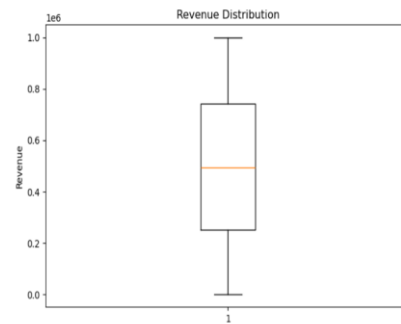
```
Data columns (total 8 columns):
```

#	Column	Non-Null Count	Dtype
0	Location	5989 non-null	object
1	Country	5989 non-null	object
2	Category	5989 non-null	object
3	Visitors	5989 non-null	int64

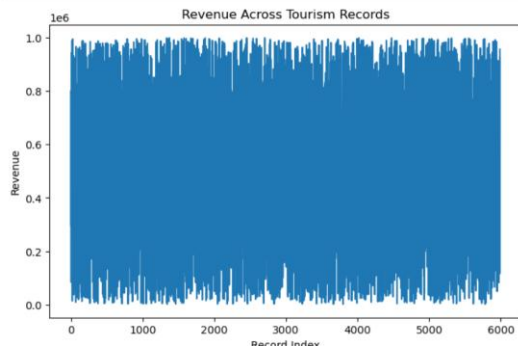
```
plt.title("Total Visitors by Tourism Category")
plt.xlabel("Tourism Category")
plt.ylabel("Total Visitors")
plt.xticks(rotations=45)
plt.show()
```



```
plt.figure(figsize=(8, 5))
plt.boxplot(df["Revenue"])
plt.title("Revenue Distribution")
plt.ylabel("Revenue")
plt.show()
```



```
plt.figure(figsize=(8, 5))
plt.plot(df.index, df["Revenue"])
plt.title("Revenue Across Tourism Records")
plt.xlabel("Record Index")
plt.ylabel("Revenue")
plt.show()
```



```
df.to_csv("tourism_dataset_cleaned.csv", index=False)

print("Cleaned dataset exported successfully!")
```

Cleaned dataset exported successfully!

RESULT: Thus, the Tourism Dataset was successfully imported, cleaned, transformed, inspected, statistically analyzed, correlated, and visualized using Python. Various Matplotlib visualizations were generated to understand visitor and revenue patterns, and the cleaned dataset was successfully exported as a CSV file.